



2.8.4 Wrap-Up: Growth Mindset

1. Check the description that matches your answer to each question for today's lesson.

Mindset Question	Best Description for Me Today
Did I use the partner discussions as learning opportunities?	○ Never ○ Sometimes ○ Usually ○ Always
Did I work on problems that challenged me?	○ Never ○ Sometimes ○ Usually ○ Always
Did I use strategies to 'un-stick' myself when I found the problems difficult?	○ Never ○ Sometimes ○ Usually ○ Always
Did I put as much effort as I possibly could into the problems?	○ Never ○ Sometimes ○ Usually ○ Always

Unit 2, Lesson 9: Dividing Polynomials



Benchmark

MA.912.AR.1.4 Divide a polynomial expression by a monomial expression with rational number coefficients.



Learning Target

- ☐ I can divide a polynomial expression by a monomial expression.



2.9.1 Warm-Up: Bell Work

Simplify each expression.

1. $1.3u(4u^5 + 7t)$

2. $\left(\frac{7}{10}g^3 + \frac{12}{5}g^2\right) + (35g^4 + 6g^2 - 10g)$

3. $\left(7x^3y - \frac{49}{10}xy\right) - \left(-\frac{11}{10}x^3y + \frac{3}{5}xy + 7xy^2\right)$



2.9.2 Guided Instruction: Dividing Polynomials by Monomials

Multiplication and division are inverse operations. A polynomial can be multiplied by a monomial. Therefore, it can also be divided by one.

1. Consider the expression $7m(m + 7)$.

a. Determine the product of $7m(m + 7)$.

b. Write two division equations using the factors $7m$ and $(m + 7)$, and the product of $7m(m + 7)$ found in Part A.

$$\underline{\hspace{2cm}} \div \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$$

$$\underline{\hspace{2cm}} \div \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$$

2. Label each part of the mathematical statement $(12x^4 - 16x^3) \div 4x^2 = 3x^2 - 4x$ using words from the bank.

Word Bank

divisor

dividend

quotient

$4x^2$	$3x^2 - 4x$	$12x^4 - 16x^3$

3. Consider the polynomial $6p^4 - 8p^3 + 2p^2$.

- a. Determine the quotient of each term of the polynomial $6p^4 - 8p^3 + 2p^2$ and $2p^2$, where $p \neq 0$.

$$6p^4 \div 2p^2 = \underline{\hspace{2cm}} \quad -8p^3 \div 2p^2 = \underline{\hspace{2cm}} \quad 2p^2 \div 2p^2 = \underline{\hspace{2cm}}$$

- b. Use the quotients found in part a to determine the quotient of $(6p^4 - 8p^3 + 2p^2) \div 2p^2$, where $p \neq 0$ using the area model.

$2p^2$	$6p^4$	$-8p^3$	$2p^2$

$$(6p^4 - 8p^3 + 2p^2) \div 2p^2 = \underline{\hspace{4cm}}$$

- c. Discuss with your partner if the quotient is a polynomial.

4. Division of a polynomial can be represented as a fraction.

- a. The division of a polynomial by a monomial is shown, where $p \neq 0$. Simplify each term by using the quotient of powers law.

$$\frac{(6p^4 - 8p^3 + 2p^2)}{2p^2} = \frac{6p^4}{2p^2} - \frac{8p^3}{2p^2} + \frac{2p^2}{2p^2} = \underline{\hspace{2cm}} - \underline{\hspace{2cm}} + \underline{\hspace{2cm}}$$

- b. Is the quotient a polynomial? Explain why or why not.

- c. Why was it stated that $p \neq 0$?

5. Consider the expression $\frac{18b^4d + 12b^3d^2 - 24bd^3}{0.6b^2d}$.

- a. Fill in the blanks in each step to determine the quotient, where $b \neq 0$ and $d \neq 0$.

$$\frac{18b^4d + 12b^3d^2 - 24bd^3}{0.6b^2d} = \frac{18b^4d}{0.6b^2d} + \frac{\quad}{0.6b^2d} - \frac{24bd^3}{\quad}$$

$$\frac{18b^4d}{0.6b^2d} + \frac{\quad}{0.6b^2d} - \frac{24bd^3}{\quad}$$

$$\begin{array}{ccc} \downarrow & \downarrow & \downarrow \\ 30b^2 & + \quad & - \quad \end{array}$$

- b. Discuss with your partner why the quotient is not a polynomial. Summarize your discussion.

6. Complete that statement.

The result of

- ☐ adding
- ☐ subtracting
- ☐ multiplying
- ☐ dividing

a polynomial

- ☐ by
- ☐ from
- ☐ to

a polynomial will not always result in a polynomial.



2.9.3 Guided Practice: Dividing Polynomials by Monomials

1. Use the area model to find $(3ac + 2.1a^3c - 1.5ac^2) \div 3ac$, where $a \neq 0$ and $c \neq 0$.

2. Find the quotient if $x \neq 0$ and $y \neq 0$.

$$\frac{56x^2y^2 - 16x^2y^4 - 48x^4y^3}{-8x^2y^4}$$

3. Divide $\left(\frac{4}{5}k^2 - \frac{2}{3}k\right) \div \left(-\frac{5}{6}k\right)$, where $k \neq 0$.



2.9.4 Your Turn!

Find each quotient.

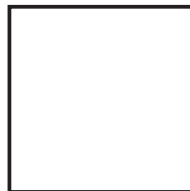
1. $\frac{7h^7 - 91h^8}{7h^7}$, where $h \neq 0$

2. $\left(\frac{1}{24}f^3r^2 - \frac{1}{4}f^4r - \frac{1}{2}f^2r\right) \div (-4f^2r)$, where $f \neq 0$ and $r \neq 0$

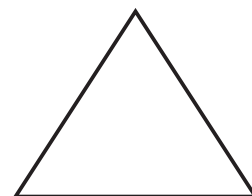


2.9.5 Wrap-Up: Reflection

1. What is something that is now “squared away” in your mind about adding, subtracting, multiplying, and dividing polynomials?



2. What are three “points” that are important about adding, subtracting, multiplying, and dividing polynomials?



3. What is something that is still “circling” your brain about adding, subtracting, multiplying, and dividing polynomials?

